

In-Class Quiz: May 12

Points: 8/8

✓ **Correct** 1/1 Points

1. which of the following memory cell types requires the least amount of physical area? 

- ☐ Master Slave D flip flop
- ☐ SRAM cell
- ☒ DRAM cell
- ☐ Both SRAM and DRAM require the same area

✓ **Correct** 1/1 Points

2. Based on the trade-offs discussed in the class, what happens if you try to reduce memory access latency? 

- ☐ Area decreases and power decreases
- ☒ Area increases and power increases
- ☐ Area decreases and power increases
- ☐ Area increases and power decreases

✓ **Correct** 1/1 Points

3. Which of the following is NOT listed as a factor that memory access latency depends on?



- ☐ Size of the memory
- ☐ Technology used (e.g., SRAM, DRAM)
- ☒ CPU clock speed
- ☐ Number of ports

✓ **Correct** 1/1 Points

4. How does the document define "Spatial Locality"? 

- ☐ If a resource is accessed, it will likely be accessed again in a short period of time.
- ☒ If a resource is accessed, similar resources will most likely be accessed again in the near future.
- ☐ The distance from the top of the memory stack to where an element is found.
- ☐ The physical distance between the L1 cache and the main memory.

✓ **Correct** 1/1 Points

5. When evaluating temporal locality in programs using a stack, what does the "stack distance" represent? 

- ☐ The total size of the stack in bytes.
- ☐ The difference between the memory addresses of the last two accesses.
- ☒ The distance from the top of the stack to where the memory address was found.
- ☐ The number of programs currently running on the processor.

✓ **Correct** 1/1 Points

6. "Address distance" (the difference between a current memory address and the closest address in the recent memory accesses) is used to demonstrate which concept? 

- ☒ Spatial Locality

- ☐ Temporal Locality
- ☐ Memory overlapping
- ☐ Latency trade-offs

✓ **Correct** 1/1 Points

7. In a typical cache hierarchy, what type of memory cells are generally used to build the L1 and L2 caches? 

- ☐ DRAM cells
- ☒ SRAM cells
- ☐ Master Slave D flip flops
- ☐ A mix of DRAM and flip flops

✓ **Correct** 1/1 Points

8. In a hierarchical memory system, what type of memory cell is typically used for Main Memory? 

- ☐ SRAM cells
- ☐ Master Slave D flip flops
- ☒ DRAM cells
- ☐ Flash memory cells

